

Aperture Tool

Software User Manual

1. Introduction

1.1 Purpose

The Aperture Tool is designed as an application/plugin for MITM (More Image Than Memory) Viewer. It interfaces directly with MITM Viewer so that the user can select, modify, import, create, and switch between various geometries on screen and see the corresponding spatial frequency domain seamlessly. This MITM Viewer application then provides an interactive platform for SAR image analysis through spatial frequency domain manipulation and sub-aperture processing. Enabling the users to perform ROI (Region Of Interest) analysis of geometries (polygon that can contain other polygon voids) by combining geometric selection in the image domain with targeted k-space filtering and reconstruction.

1.2 Document Overview

This Software User Manual (SUM) is structured to provide users with all the necessary information to install and effectively use the RCS tool. The manual is organized as follows:

1. **Introduction:** An overview of the software's purpose and design
2. **Installation Instructions:** Step-by-step guidance for setting up the software
3. **Graphical User Interface:** Detailed explanation of the interface and navigation
4. **Model View Controller:** Technical overview of the software design pattern used
5. **Conclusion:** Documentation summary
6. **References:** Citations for references.

2. Installation Instructions

Python Install

1. Navigate to the main rcs-tool directory
2. `pip install -r requirements.txt`
3. `pip install src/.`

4. Install PyMITM

Standalone Application

PyRCS can be packaged as a standalone executable using PyInstaller:

1. Install PyMITM if not already installed:

2. Navigate to the PyInstaller directory

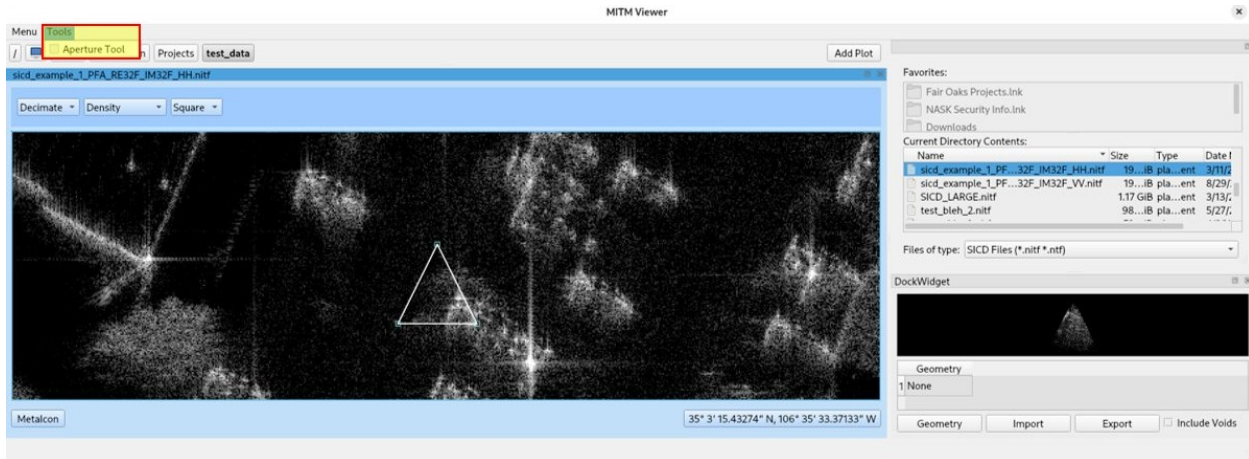
3. Create the executable:

```
pyinstaller mitm.spec
```

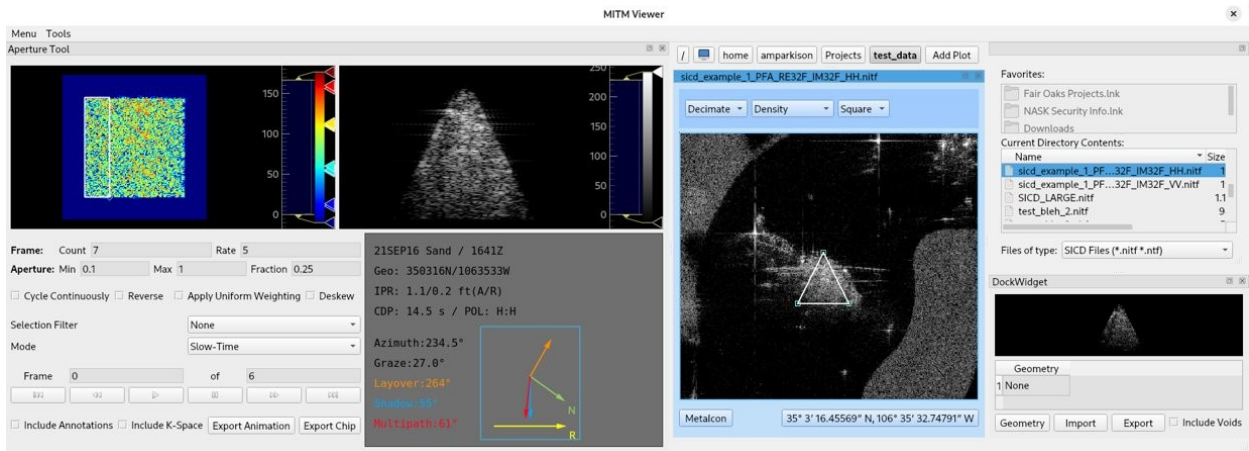
4. Run the executable:

```
./dist/mitm.exe
```

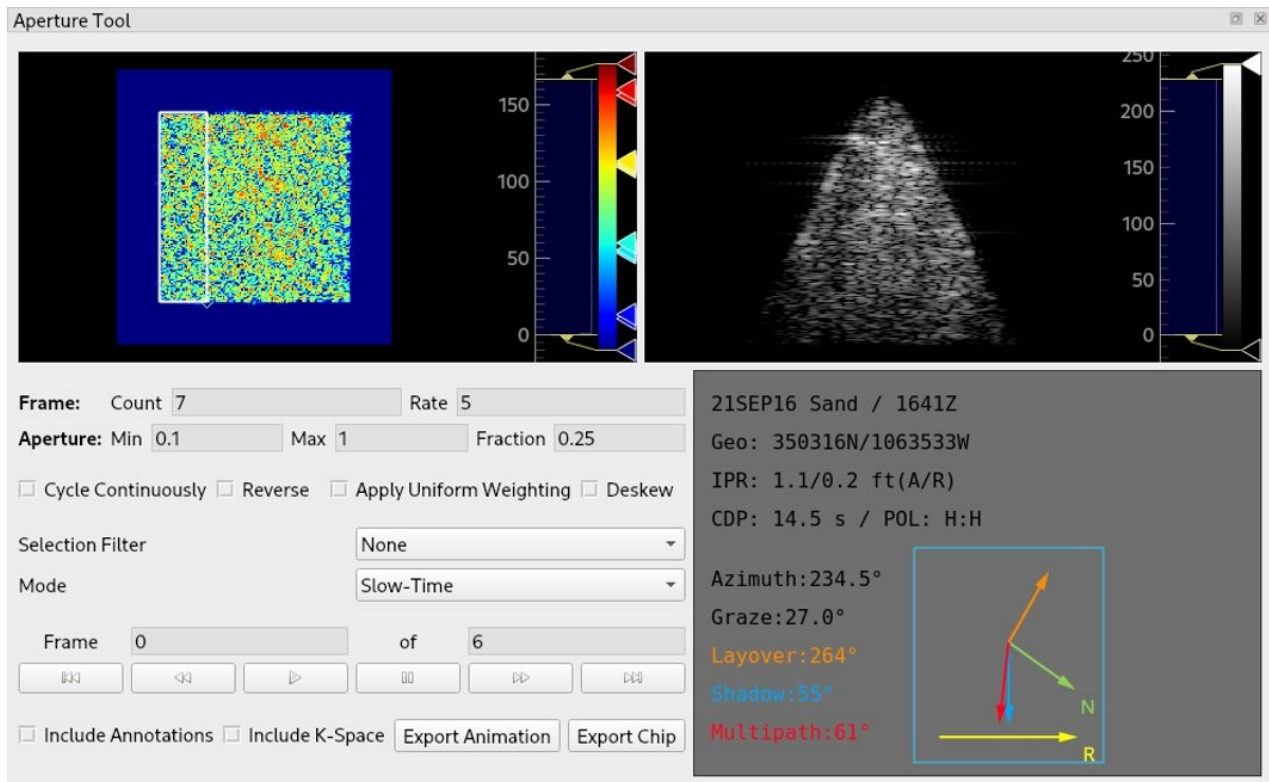
3. Graphical User Interface (GUI)



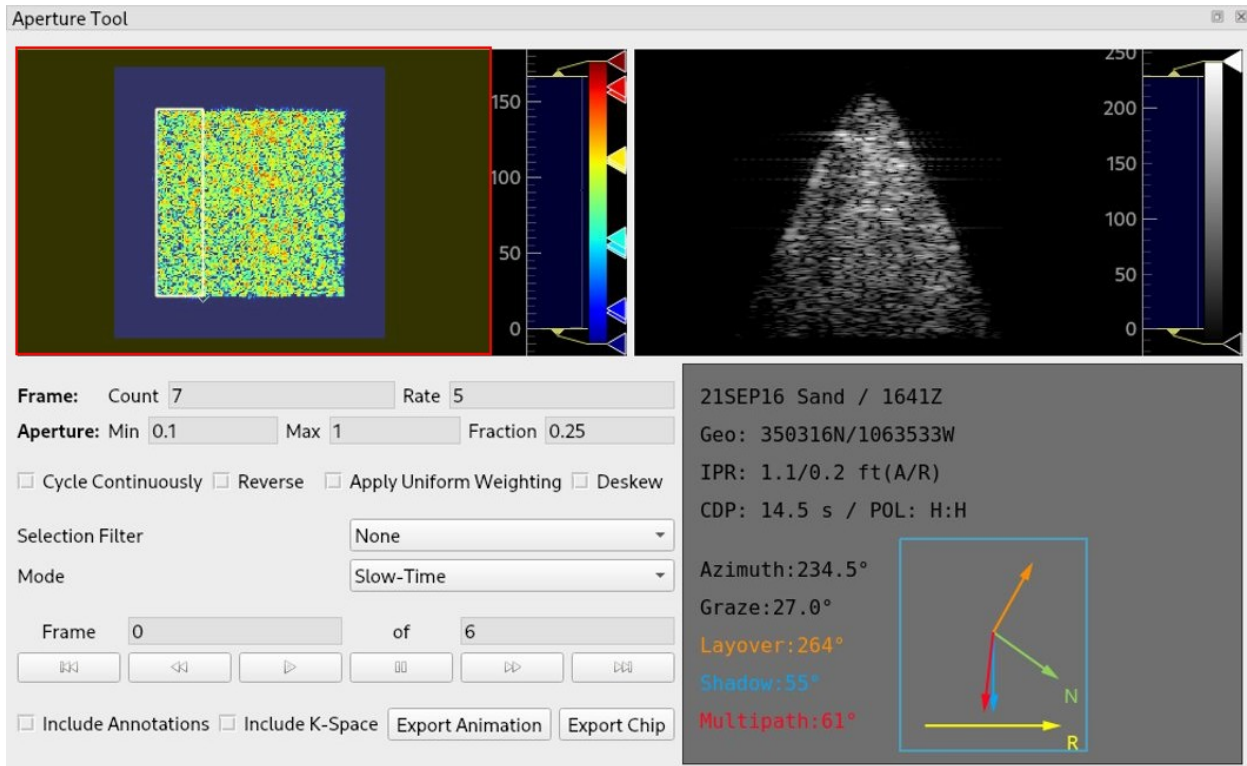
The Aperture Tool GUI exists as one main dock widget that gets added to MITM Viewer, by default this is displayed to the left of MITM Viewer but can be moved around and docked and undocked. To activate the Aperture Tool dock widget the user can go to the tools menu in MITM Viewer and check the “Aperture Tool”.



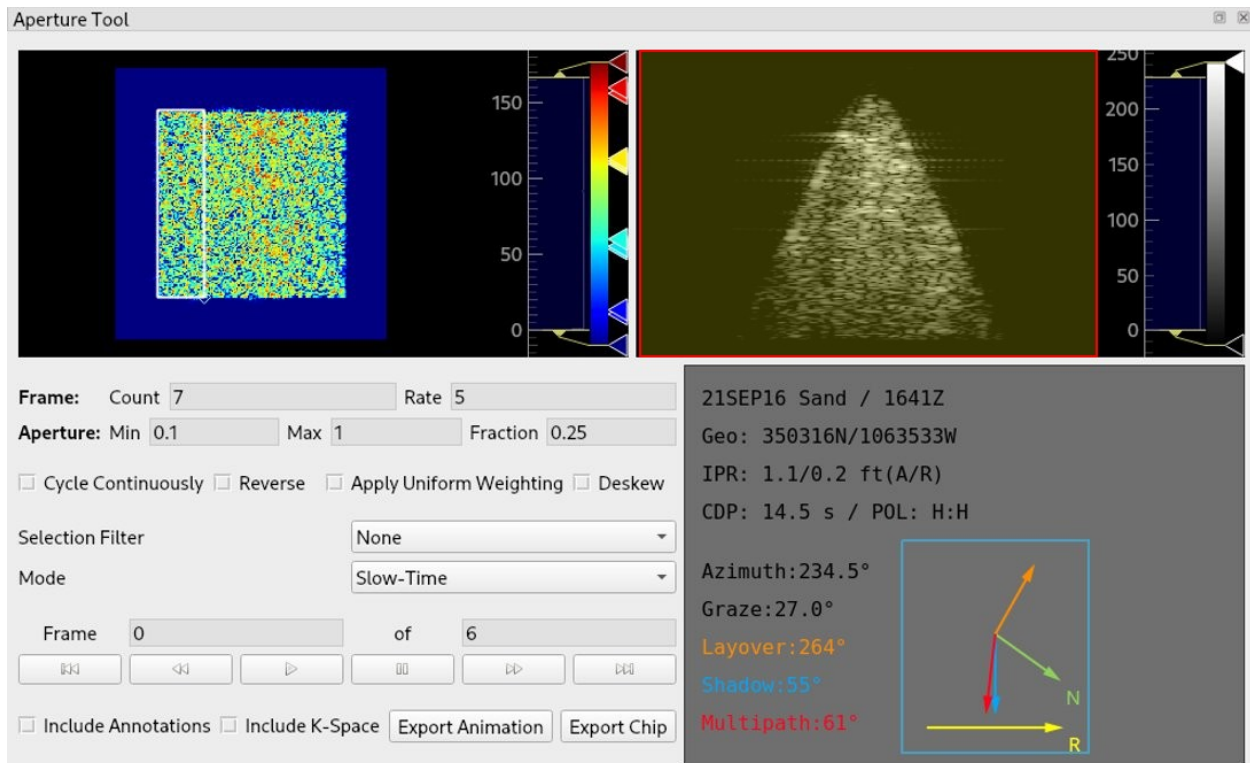
The Aperture Tool dock widget is displayed on the left.



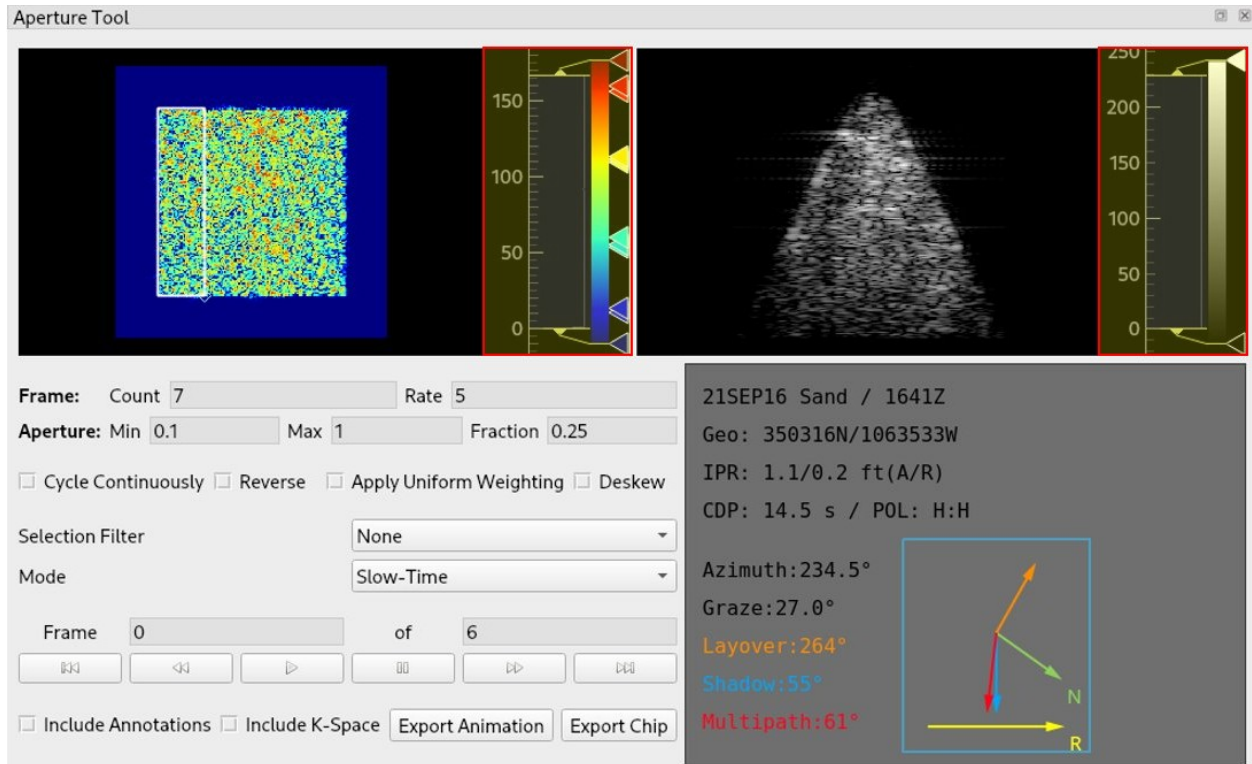
The user can click on different geometries or modify currently selected geometries to update the reference geometry that the Aperture Tool is currently using for processing.



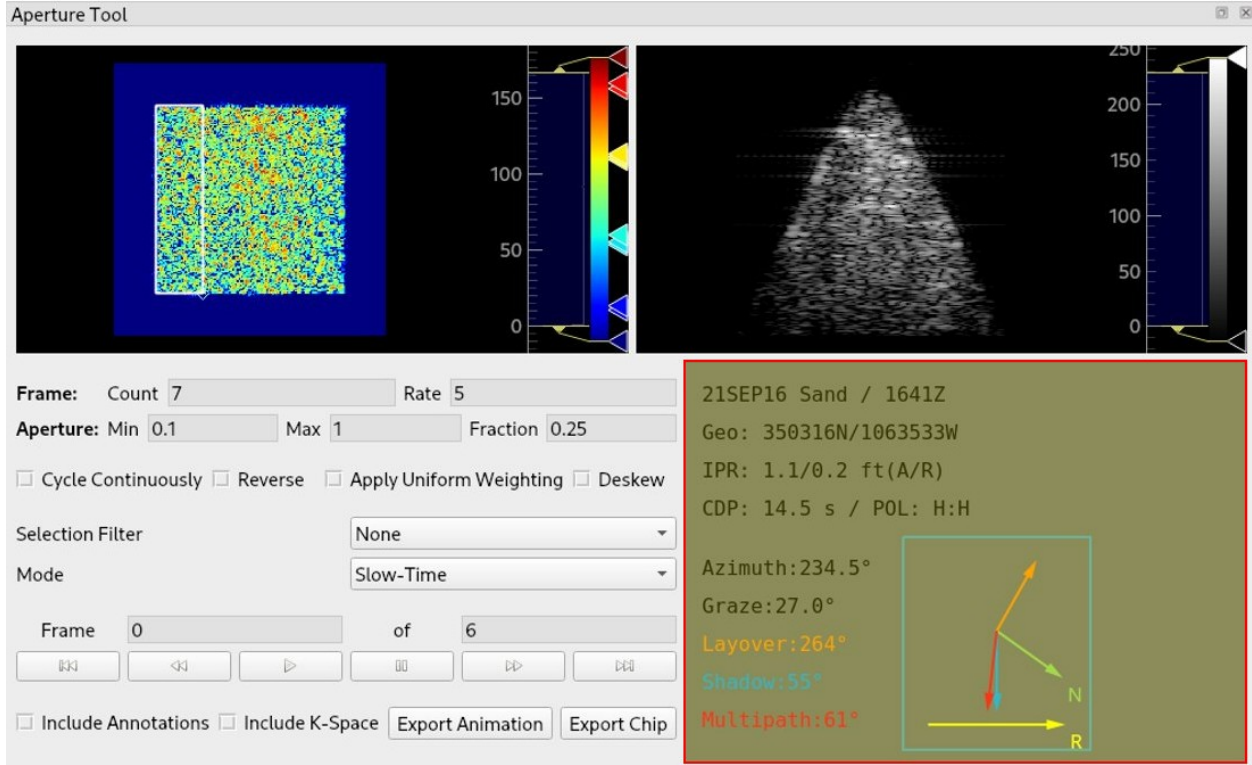
The top left display shows the k-space (spatial frequency domain) of the currently selected geometry, by default a sub-aperture ROI (region of interest) geometry is spawned in on this k-space. This ROI is constrained to the Fourier transform bounds. The user can modify the size of this ROI by selecting the bottom right hand corner (diamond handle) and dragging it. The user can also drag the ROI around to different regions in the k-space.



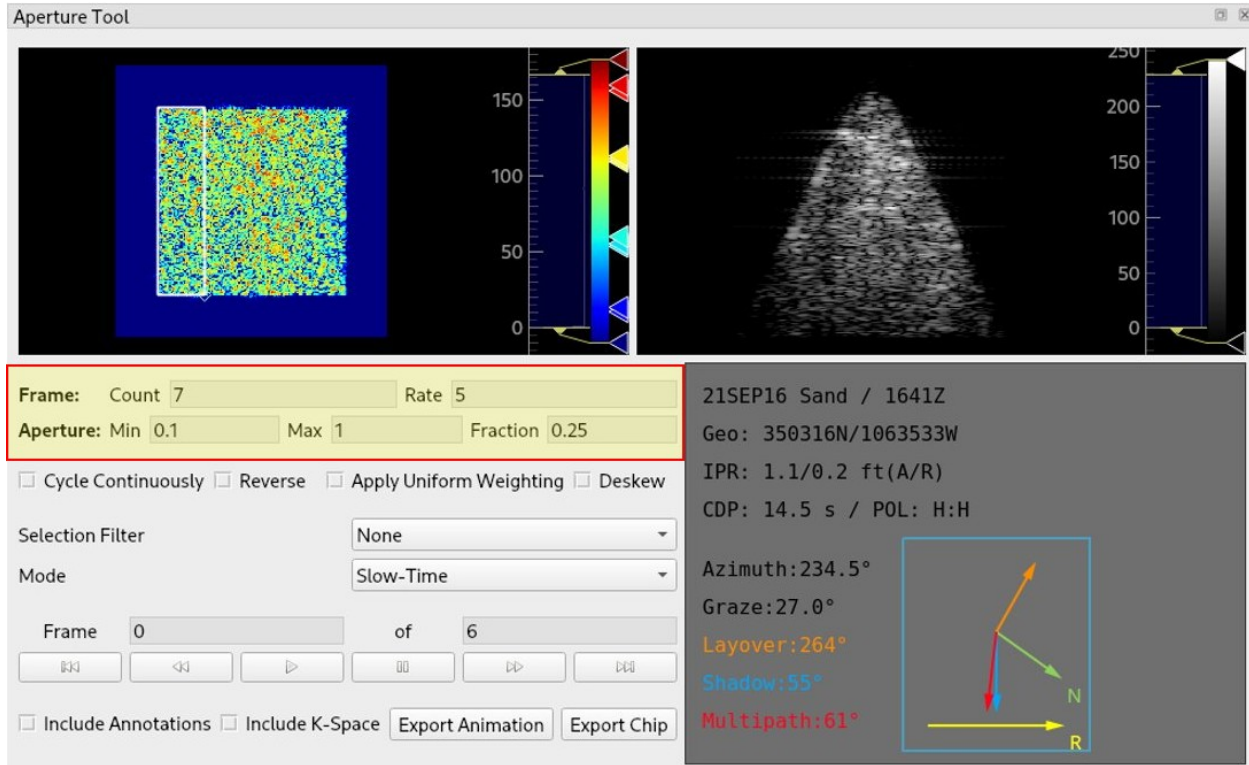
Any changes to this ROI will automatically update the filtered view on top right display. This shows the post processed sub-aperture image view.



To the right of each display is a histogram widget allowing the user to modify the color gradient for better data visualization.



The bottom right corner display shows the meta icon. This will update for each frame.



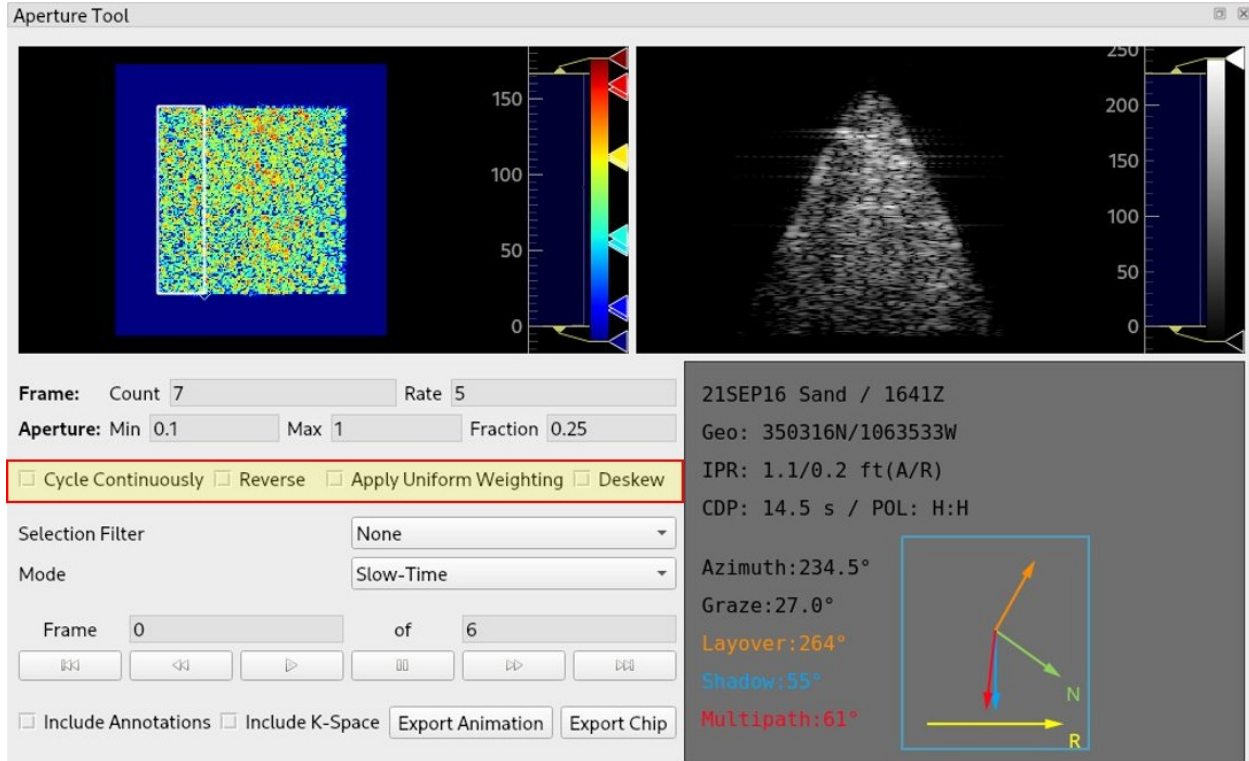
The top portion of the main layout consists of editable fields for modifying the frame count, frame rate, aperture min, aperture max, and aperture fraction. These are main parameters controlling the animations that will be created.

Frame Count: Total number of frames in the animation.

Frame Rate: Playback rate of animation.

Min/Max Aperture: Range limits for aperture percentage animations.

Aperture Fraction: Percentage of total aperture to use in animations.



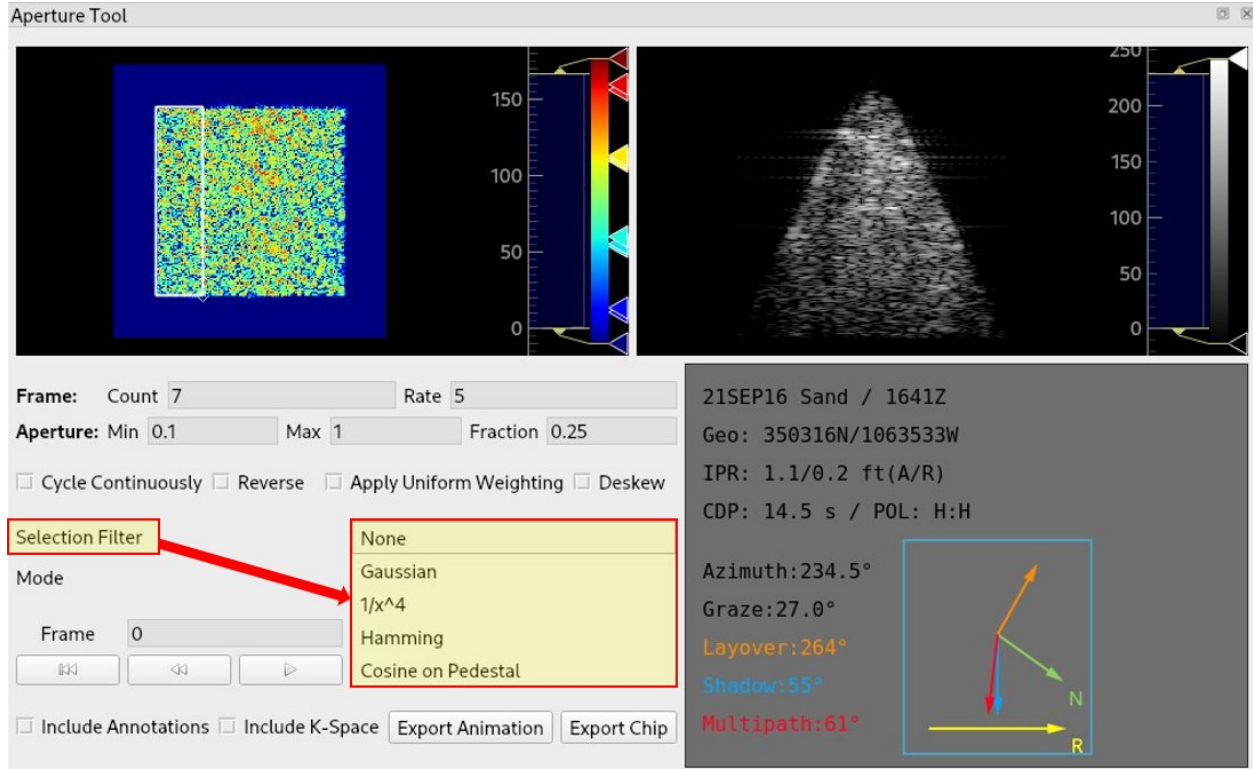
This portion of the GUI has the two option for animation playback and then an option for applying uniform weighting and deskew to the data. These last two options are only accessible to the user if the users data has the required meta data to support them.

Cycle Continuously: Toggle animation playback to loop continuously.

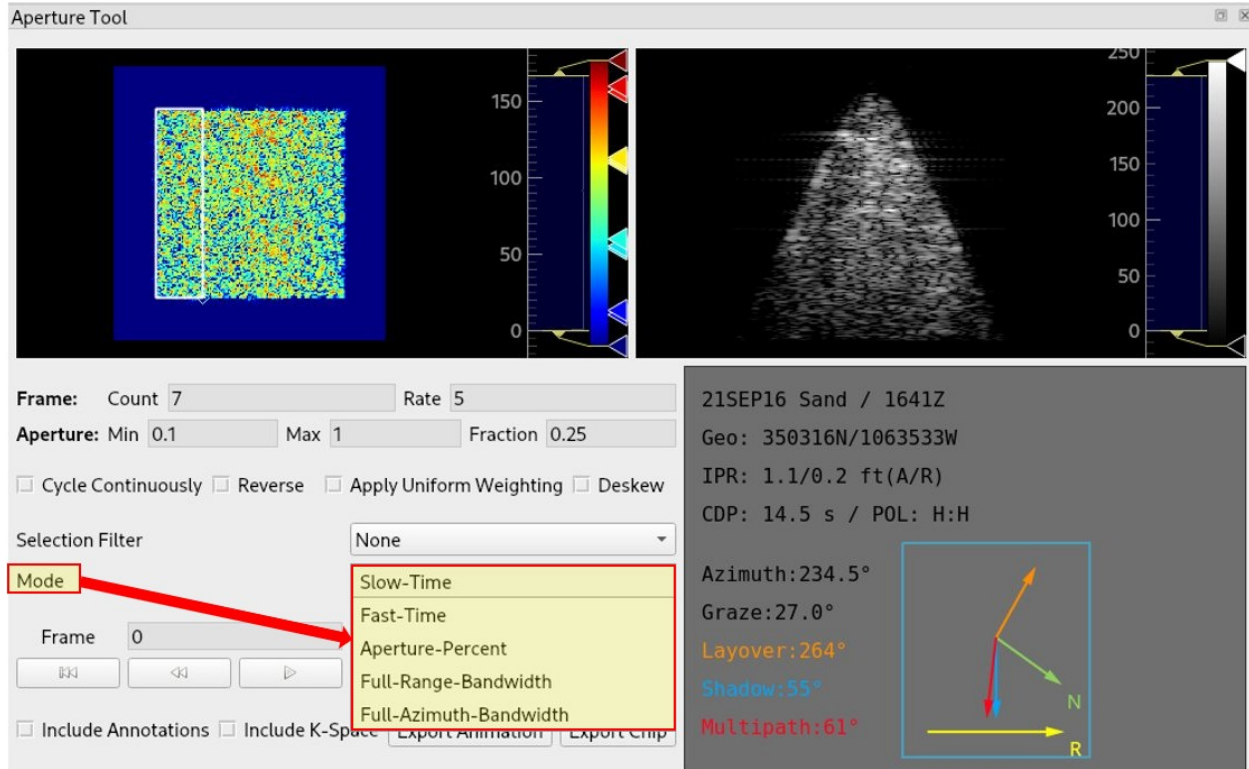
Reverse: Toggle between forward or backward animation playback.

Apply Uniform Weighting: Toggle removal of amplitude tapering applied during data collection if available.

Deskew: Toggle removable of linear phase ramps to improve image focus if available.



The **Selection Filter** drop-down menu allows the user to apply a windowing function to the data. The current supported window functions are **Gaussian**, **$1/x^4$** , **Hamming**, and **Cosine on Pedestal**.



The **Mode** drop-down menu allows the user to switch between different animation modes.

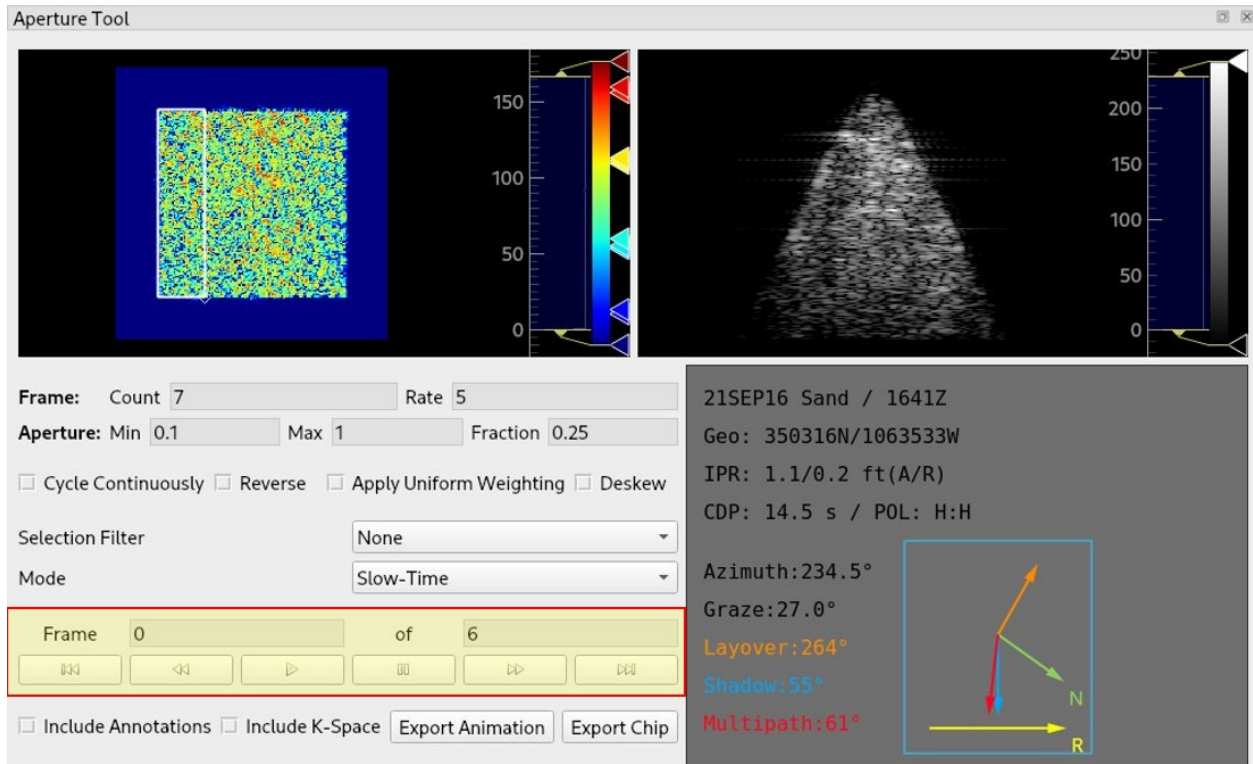
Slow-Time Animation: Shows aperture movement in the cross-range (azimuth) direction

Fast-Time Animation: Shows aperture movement in the range direction.

Aperture Percentage: Gradually changes the size of the processed aperture with bounds defined by the aperture min and max

Range Bandwidth: Varies the range frequency content

Azimuth Bandwidth: Varies the azimuth frequency content



This portion of the GUI contains actual animation controls, as well as displaying the current animation frame and the total number of frames. Below are the button functions going from right to left sequentially.

Step Start: Steps to the first frame in the animation.

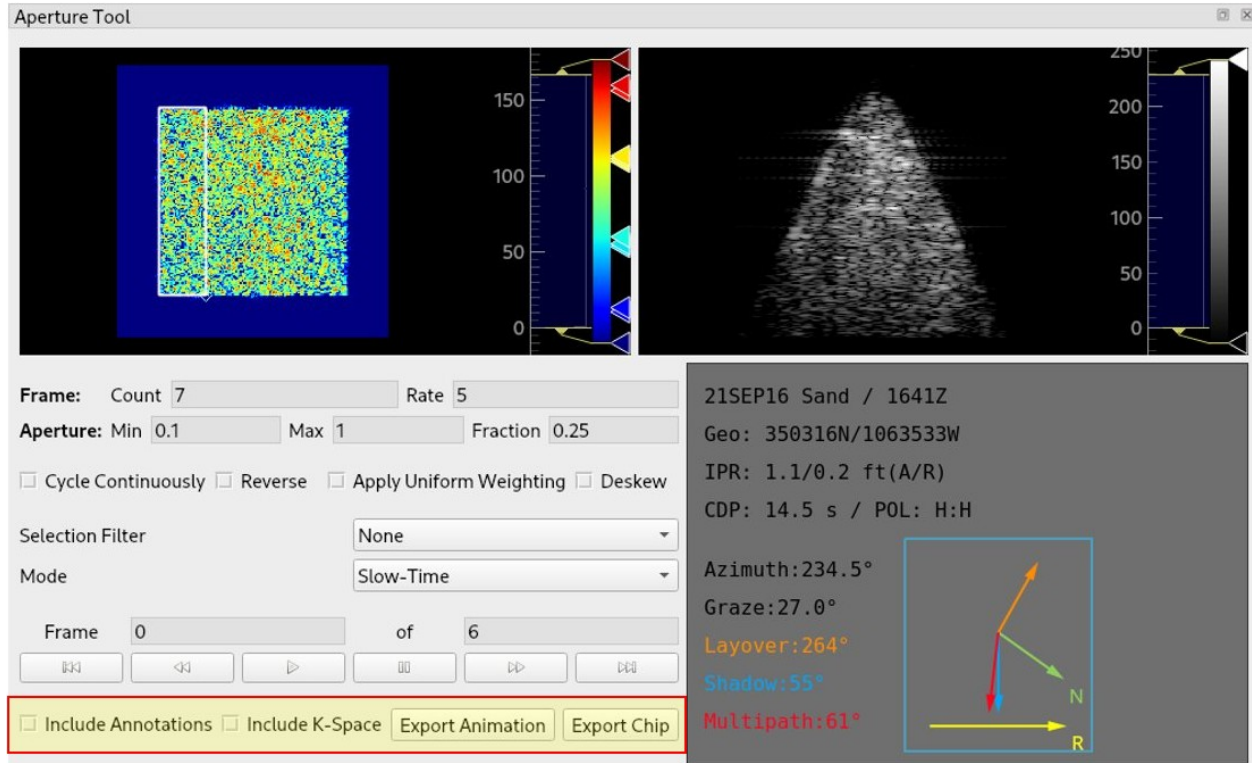
Step Back: Steps back one frame in the animation.

Play: Plays the animation.

Pause: Stops the animation.

Step Forward: Steps forward one frame in the animation.

Step End: Steps to the last frame in the animation



This portion of the GUI contains options pertaining to exporting.

Toggle Include Annotations: Toggles the inclusion of a frame count display on bottom left corner of the K-space portion of the animation.

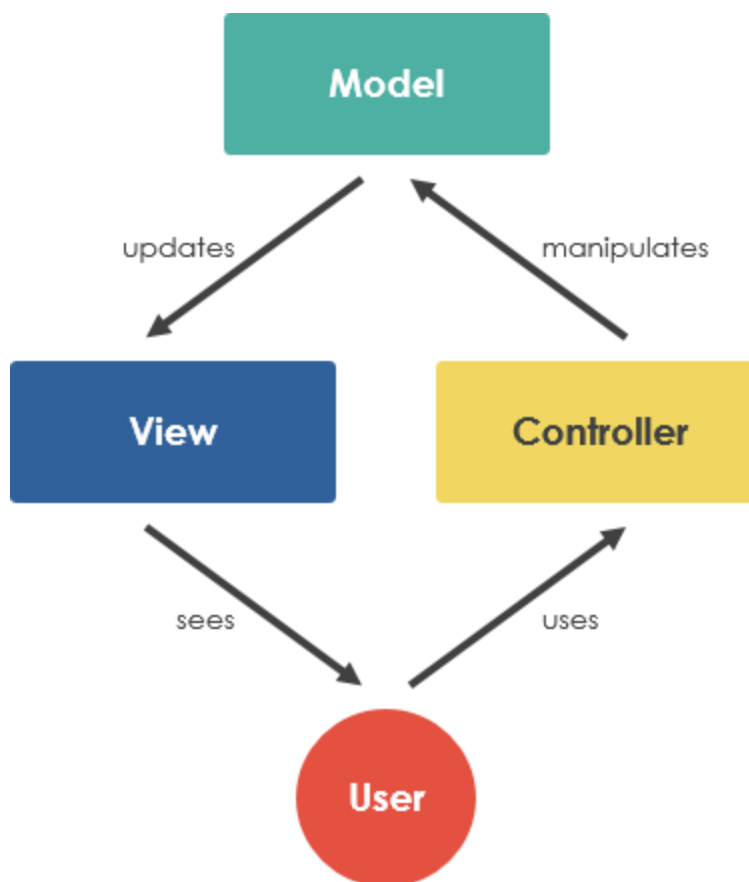
Toggle Include K-Space: Toggles the inclusion of a side by side view of the K-space and filtered view in the animation.

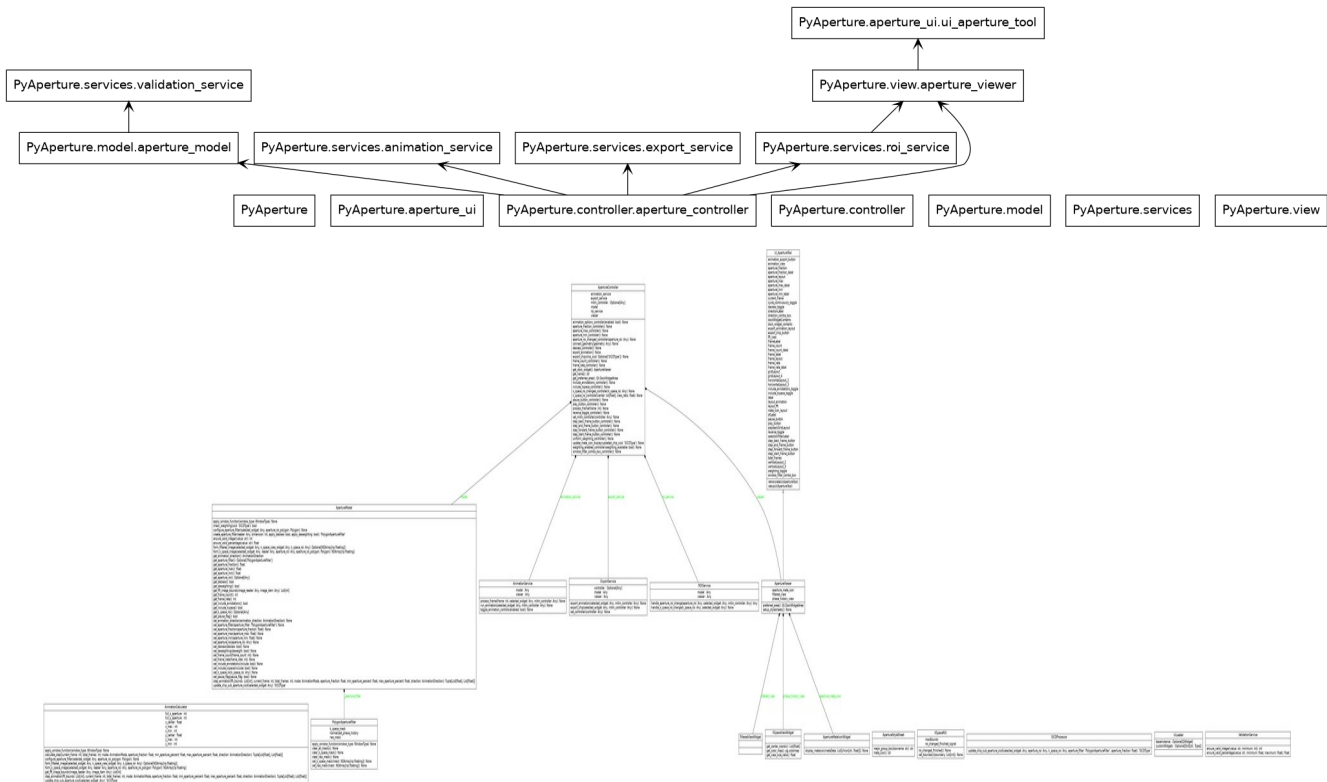
Export Animation: Exports animation as a mp4 with the settings toggled above.

Export Chip: Exports the resulting sub-aperture chip as a NITF.

4. Model View Controller (MVC)

The Aperture Tool was made following the MVC (Model-View-Controller) design pattern. MVC separates the application in three main components. The model and viewer are completed isolated from one another, the model contains all of the brains of the application and functionality, the viewer contains all of the GUI elements and handles user interaction. The controller marries the model and viewer so that as a user interacts with the GUI any back-end calculations or logic can be completed by the model by having the controller send the input data from the viewer to the model and then back to the viewer to update the information to the user.





5. Conclusion

In conclusion the Aperture tool gives users the ability perform sub-aperture processing and k-space analysis in an integrated environment allowing seamless integration with other tools.

6. References

[1] Visual Paradigm. From Use Case to MVC Framework: A Guide Object-Oriented System Development. [Online]. Available: <https://guides.visual-paradigm.com/from-use-case-to-mvc-framework-a-guide-object-oriented-system-development/>. [Accessed: March 31, 2025].